LOAN APPROVAL DATASET

IA – 1 Assignment

Team 9

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Loan Approval using ML Algorithm

I. INTRODUCTION

Machine learning is a subfield of artificial intelligence (AI) that deals with the development of algorithms and statistical models that enable machines to learn from data and make predictions or decisions without being explicitly programmed.

Machine learning is a way of enabling computers to automatically improve their performance on a particular task by learning from experience or data. It also involves the use of algorithms and statistical models to analyze and make predictions or decisions based on patterns in the data.

Machine learning can be broadly categorized into three types: supervised learning, unsupervised learning, and reinforcement learning.

Supervised learning involves training a machine learning model using labelled data, while **unsupervised learning** involves training a model without labelled data. **Reinforcement learning**, on the other hand, involves training a model using a reward-based system, where the model learns by trial and error.

Machine learning has a wide range of applications, including image and speech recognition, natural language processing, recommendation systems, fraud detection, and autonomous vehicles, among others.

Data mining techniques are very useful in this field. Data mining is a process of extracting meaningful knowledge from a large and unstructured amount of data. Such extracted information can help in decision-making through performing important operations.

For the loan approval prediction, there are many traditional data mining methods. Machine-learning algorithms are available to automate the loan status prediction system. Which are used to predict the right person who deserves the loan, many machine learning algorithms, such as decision tree, gradient boosting, random forest, logistic regression, and many more, can predict the right person who deserves the loan. Therefore, this paper aims at risk to approve the loan by predicting the loan status based on different loan factors or features.

The purpose of the loan can be anything based on the customer's needs. Loans are broadly divided as open ended and close-ended loans. Examples of open-end loans are credit cards and a home equity line of credit (HELOC).

Close-ended loans decreases with each payment. It means the amount is reduced after an instalment. In other words, it is a legal term that cannot be modified by the borrower. Personal loans, mortgages, auto payments, EMI and student loans are the most common examples of close-ended loans. Secured or collateral loans are those loans that are protected by an asset. Houses, Vehicles, Savings accounts are the personal properties used to secure the loan.

Distribution and approval of loans is one of the core business parts of any operational retail bank. It resembles a major income for the bank's capital for its assets. With the enhancement of the technologies in the banking sector and the daily processes becoming much more streamlined and efficient, this caused a larger influx of applicants for loans seeking capital from banks which readily expands any bank reservoir of data for this type of decision making. The process of selecting an eligible candidate is a typical process and each financial institution is somewhat unique and each follows a specific standardized workflow. This type of decision making has an inherent risk associated with it. Moreover, selecting and developing the proper resources to make that judgment also possess another operational risk and cost.

Loan Prediction is very helpful for employee of banks as well as for the applicant also. The aim of this Paper is to provide quick, immediate and easy way to choose the deserving applicants. It can provide special advantages to the bank. The Loan Prediction System can can automatically calculate the weight of each features taking part in loan processing and on new test data same features are processed with respect to their associated weight .A time limit can be set for the applicant to check whether his/her loan can be sanctioned or not. Loan Prediction System allows jumping to specific application so that it can be check on priority basis. This Paper is exclusively for the managing authority of Bank/finance company, whole process of prediction is done privately no stakeholders would be able to alter the processing. Result against particular Loan Id can be send to various department of banks so that they can take appropriate action on application. This helps all others department to carried out other formalities.

II. LITERATURE REVIEW

By using the data, we need to build the model that will help to predict the to give the loan to that person depending on his information like gender, married or not, dependent or not, graduation, self-employed, applicant income, Co-applicant income, loan amount etc. There are so many models which can predict the loan approvals.

Suliman Mohamed Fati studied at College of Computer and Information Sciences, Prince Sultan University. Riyadh, Saudi Arabia. He did Machine Learning-Based Prediction Model for Loan Approval by Appling the three machine learning algorithms Logistic Regression, Decision Tree and Random Forest. The process he did is firstly the data source he collected then done some dataset Description and Pre-processing. Decided which model is suitable for this data source. Then he applied the suitable machine learning algorithms.

Ashwini S. Kadam1, Shraddha R. Nikam2, Ankita A. Aher3, Gayatri V. Shelke4, Amar S. Chandgude 5 written the paper on the Prediction for Loan Approval Using Machine Learning Algorithm. Here they used Naïve Bayes (NB) Model and SV method. The approached steps for this First they did Exploratory Data analysis and then the Machine Learning Steps Data Collection, Data Cleaning and pre-processing, Model selection, data verification, classification and Report deliver.

Kumar Arun, Garg Ishan and Kaur Sanmeet these are from Thapar University, India. They written the paper how to approach to the loan approval prediction based on machine learning. They used machine learning algorithms like Decision Tree, Random Forest, Support Vector Machine, Linear Models, Neural Network, and Adaboost. They give the basic approach and the suitable algorithms which we can apply for the Loan Approval Problem.

Mohammad Ahmad Sheikh he is the student at Galgotias University Greater Noida, India. He and another two professors from the same university and the department of school of computer science and engineering written a paper "An Approach for Prediction of Loan Approval using Machine Learning Algorithm". They came with the problem statement.

Loan prediction is a very common real-life problem that every finance company faces in their lending operations. If the loan approval process is automated, it can save a lot of man hours and

improve the speed of service to the customers. The increase in customer satisfaction and savings in operational costs are significant.

However, the benefits can only be reaped if the bank has a robust model to accurately predict which customer's loan it should approve and which to reject, in order to minimize the risk of loan default.

They did a neat approach to this problem statement they used the logistic regression model for this analysis. They did some model evaluation techniques like Confusion metrics, Accuracy, Precision, Recall and F1 score. These are the techniques which help to evaluate the performance of the model.

Some of the faculties of engineering and computer sciences, iNurture, TMU, Moradabad and another from Department of Computer Science, Government Engineering College, Bharatpur. The name of the faculties is Anshika Gupta, Vinay Pant, Sudhanshu Kumar and Pravesh Kumar Bansal written a paper on the Bank Loan Prediction System using Machine Learning. In this first they prepared the methodology regarding that they used logistic regression and random forest machine learning algorithms for the loan prediction.

Tree-Based Methods for Loan Approval written by Mohamed Alaradi pursuing MSc. in Big Data Science and Analytics University of Bahrain. He applied some techniques like Decision tree, Bagging, Random Forest, Boosting, Performance Metrics, programming software. By applying they got to know about the accuracy of each method which will help in the Loan Approval process.

Loan Prediction is very helpful for employee of banks as well as for the applicant also. The aim of this Paper is to provide quick, immediate and easy way to choose the deserving applicants. It can provide special advantages to the bank. The Loan Prediction System can automatically calculate the weight of each features taking part in loan processing and on new test data same features are processed with respect to their associated weight.

There are numerous fields where data mining can be used in financial industries such as customer segmentation and profitability, applicants for high-risk loans, predicting default payments, promotions, collateral monitoring, rating assets, irregular sales, managing stock holdings, cash management and estimating activities, most profitable clients with credit cards and cross-selling.

One of the common tools used today in data mining is Weka. The Weka facilitates these traditional data mining functions in the pre-processing, classification, clustering, association, regression and feature selection phases. It also provides a visualization tool. And it is useful in learning basic machine learning principles with various options and in analyzing the output being generated.

The current work aims at automating the process of loan approval by using statistical learning methods based on a bank's historical data that outlines certain characteristics of the applicants. The rationale behind this work is obviously minimizing risk and enabling bank personnel to take better decision and judgment on selecting eligible candidate for loan approval.

Prediction of a modernized loan approval system is incredibly helpful for banks and also the clients. This system checks the candidate on his priority basis. Customer can submit his application directly to the bank so the bank will do the whole process, no third party or stockholder will interfere in it. And finally, the bank will decide that the candidate is deserving or not on its priority basis. The only object of this research paper is that the deserving candidate gets straight forward and quick results.

Three Machine Learning algorithms which are used to find out the correct prediction of Data set. XGBoost, Random Forest and decision tree.

P. L. Srinivasa Murthy1, G. Soma Shekar2, P. Rohith3, G. Vishnu Vardhan Reddy4 written the paper on Loan Approval Prediction System Using Machine Learning. The system is highly effective for identifying eligible loan applicants and can cater to all banking needs. It can seamlessly integrate into any operating system and has a promising future with the shift towards online platforms. The system ensures strong security and reliability, Here the author used three algorithms for prediction of loan. They are 1.) K Nearest Neighbor 2.) Decision Tree 3.) Random Forest Algorithm, which provides precise outcomes. It can handle a large number of customers without any complications, and there is potential for enhancing accuracy by incorporating additional algorithms in the future.

An Approach for Prediction of Loan Approval Using Machine Learning Algorithm written by Ms. Kathe Rutika Pramod1, SVIT, Nashik2, Ms. Dapse Punam Laxman3, Ms. Panhale Sakshi Dattatray4, Mr. Ghorpade Dinesh B.5, Ms. Avhad Pooja Prakash6. The analytical process included data cleaning, missing value imputation, exploratory analysis, and model building with an accuracy of 0.811 on the public test set. It revealed that applicants without a passing credit history are often rejected due to a higher probability of defaulting. Additionally, applicants with higher incomes but lower loan amounts are more likely to be approved as they are perceived as reliable borrowers. Surprisingly, basic characteristics such as gender and marital status do not significantly influence the company's approval decisions.

Loan Approval Prediction using Machine Learning Algorithms Approach written by Nitesh Pandey1, Ramanand Gupta2, Sagar Uniyal3, Vishal Kumar4. Here the author used three algorithms for prediction of loan. They are 1.) Support Vector Machine(SVM) 2.)Decision Tree 3.)Random Forest Algorithm 4.)Logistic Regression. This paper focused on using machine learning for loan acceptance prediction. The process involved data pre-processing, handling missing values, and conducting experimental data analysis. The precision rates of the evaluated algorithms ranged from 70% to 80%. Notably, the Support Vector Machine model demonstrated high efficiency and outperformed other models, making it the preferred choice for loan acceptance prediction.

An Approach for Prediction of Loan Approval using Machine Learning Algorithm written by Mohammad Ahmad Sheikh1, Amit Kumar Goel2, Tapas Kumar3. Here the author used algorithm for prediction of loan. That is Logistic Regression. The process of prediction starts from cleaning and processing of data, imputation of missing values, experimental analysis of data set and then model building to evaluation of model and testing on test data. On Data set, the bestcase accuracy obtained on the original data set is 0.811. The following conclusions are reached after analysis that those applicants whose credit score was worst will fail to get loan approval, due to a higher probability of not paying back the loan amount. Most of the time, those applicants who have high income and demands for lower amount of loan are more likely to get approved which makes sense, more likely to pay back their loans. Some other characteristic like gender and marital status seems not to be taken into consideration by the company.

In Somayyeh. Z et al[1] a model was proposed for predicting and identifying the right applicants who have applied for loan. So here Decision Tree technique is implemented to estimate the traits where accuracy rate is not much appreciable.

In Sudhamathy G et al ., This paper aims to build a model. It was based on decision tree where classification is used. It also uses the functions in the R-Package. Before preparing the model, the data is synchronized, made it ready to deliver effective results. The model prepared is used for predict using the dataset and the results shows the accuracy. In Dileep. B et al, Data analysis was done by figuring out techniques like Bayes classification, Decision Tree, Logistic Regression, K means algorithm, Neural Network Techniques, Perception model are combined in this model.

Kshitiz Gautam, Arun Pratap Singh, Keshav Tyagi, Mr. Suresh Kumar in Loan Prediction using Decision tree and Random Forest used training data set to train our model. After the model is trained, we test it with unknown examples from the same scenario. Another

process that we use before testing and training data is data pre-processing. In data pre-processing we remove all sorts of values that can cause an error like redundant values, incomplete values, missing data, etc.

From all the research papers we can observe that there are some common steps which are applied by every above researcher.

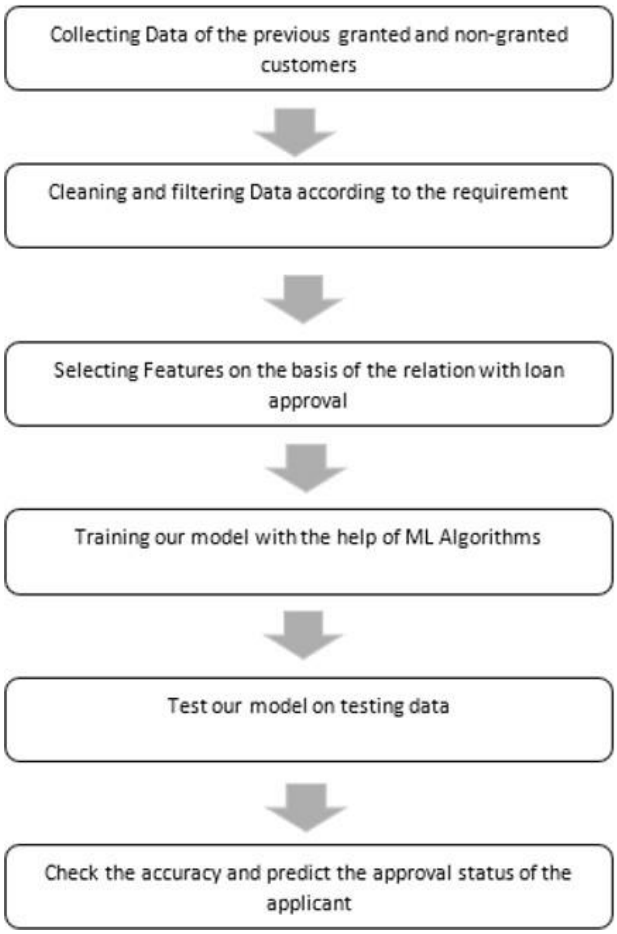

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Table 1: The Following is the Summary of the existing works on the given domain:

Year	Previous Works	Tool used	Techniques used	Advantages	Disadvantages	Ref.
2021	[1]	Python	Logistic Regression Box plot	The Cross Validation Score of 80.946% which is best value when compared to recent models based on other Machine Learning Models. This program takes the computer file it returns the output within the type of binary i.e., either 0 or 1	--------	9
2021	[2]	R	Decision Tree (C5.0), Random Forest, Support Vector Machine (SVM), linear Model, Neural Network (Nnet), Adaboost (ADB).	The product is a highly efficient component. This application works properly and meets all Banker requirements. This component can be easily plugged into many other systems.	computer glitches, errors in content and most important weight of features are fixed in automated prediction system, So in the near future the so –called software could be made more secure, reliable and dynamic weight adjustment.	10
2020	[3]	R	Decision tree, Random Forest	This application operates efficiently and fulfilling all the major requirements of Banker. Although the application is flexible with various systems and can be plugged effectively.	Numbers of cases of computer glitches, errors in content and most important weight of features are fixed in automated prediction system	7
2021	[4]	Python	XGBoost, Random Forest, Decision Tree	The advantage of this system is that we provided some conditions by setting the algorithms and just by evaluating the details, we get to know eligibility criteria that client is eligible or not.	some situations like client going through some disaster so here the algorithm cannot predict the appropriate result.	5
2020	[5].	Python	Decision Tree Bagging Random Forest Boosting	In this work, many statistical learning classification techniques were performed under the pursuit of predicting loan approval status for a bank	Numbers of cases of computer glitches, errors in content.	13
2021	[6]	Weka	Decision Tree Logistic Regression	The output is in the form of 0 and 1(binary form), were '1' indicates that loan is approved and '0' for loan rejected.	some situations like client going through some disaster so here the algorithm cannot predict the appropriate result.	8
2020	[7]	Python	Confusion Metrics, Accuracy, Precision, Recall, F1 Score	Among all the techniques, the best-case accuracy obtained on the original data set is 0.811.	Some other characteristics like gender and marital status seems not to be taken into consideration by the company	15
2021	[8]	Python	Naïve Bayes (NB) Model and Support Vector Machine (SVM)	Naïve Bayes model is extremely efficient and gives a better result when compared to other models. It works correctly and fulfills all requirements of bankers.	Support Vector machine (SVM) won't give that much accuracy and efficient result.	23
2016	[9]	R	Decision Tree (C5.0), Random Forest (RF), Support Vector Machine (SVM), Linear Models (LM), Neural Network (Nnet), Adaboost (ADB)	It can be safely concluded that the product is a highly efficient component.	There have been numbers of cases of computer glitches, errors in content and most important weight of features is fixed in automated prediction system	3
2020	[10]	Python	Logistic Regression, Random Forest, Correlation between Parameters	This model is used for the banking system or anyone who wants to apply for a loan. It will be very helpful in bank management. It reduces all the frauds done at the time of loan approval.	As it seems, it will not deal with some special cases when only one parameter is enough for the decision, but it is quite efficient and reliable in some instant.	10

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